

CodeAlpha Cyber Security Internship

PROJECT REPORT

Project Title: Basic Network Sniffer using Python & Scapy

Student Name: Muskaan

Internship: CodeAlpha Cyber Security

Certificate

This is to certify that **Muskaan** successfully completed the project '**Basic Network Sniffer using Python & Scapy**' as part of the CodeAlpha Cyber Security Internship.

Mentor Signature: _____

Acknowledgement

I express my sincere gratitude to CodeAlpha for providing the opportunity to work on this cybersecurity project. This project helped me gain practical experience in packet sniffing, network protocols, Scapy, and traffic analysis. I also thank everyone whose documentation and learning resources supported this work.

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Abstract

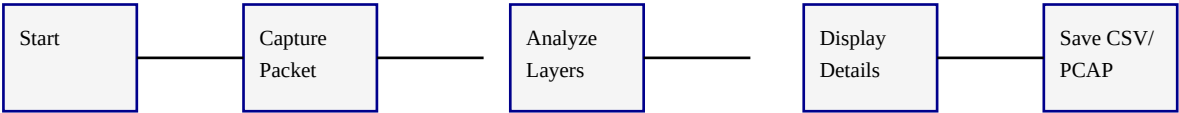
This project implements a live network packet sniffer using Python and Scapy. It captures packets, extracts Ethernet, IP, TCP/UDP/ICMP information, displays timestamps, MAC/IP addresses, ports, flags, payloads, and exports the captured traffic to CSV and PCAP formats.

Objectives

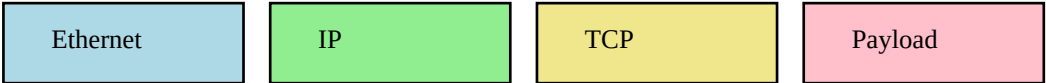
- Capture live packets
- Understand TCP/IP
- Analyze packet layers
- Export results
- Develop practical cybersecurity skills

Network Architecture Diagram

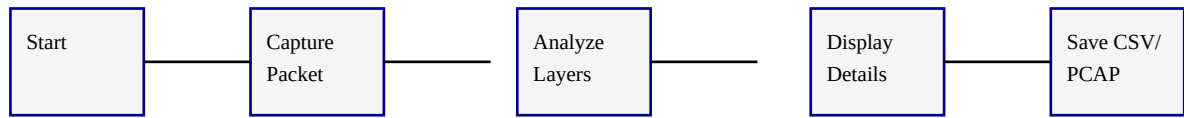
Packet flow through the network:



Packet Structure Diagram



Program Flowchart



The application starts packet capture, processes each packet, extracts required fields, displays the information, and stores it in CSV and PCAP files.

Source Code Explanation

- `sniff()` captures live packets.
- `haslayer()` checks whether a protocol layer exists.
- `packet[IP].src` and `packet[IP].dst` retrieve IP addresses.
- TCP/UDP layers provide port information.
- `wrpcap()` saves packets to a PCAP file.
- CSV logging stores packet metadata for later analysis.

Output Screenshots

```
PS E:\Code Alpha\CodeAlpha_BasicNetworkSniffer> python sniffer.py
```

```
=====
CODEALPHA BASIC NETWORK SNIFFER
=====
```

```
Capturing 20 packets...
```

```
=====
Packet #11
=====
```

```
Timestamp      : 20-07-2026 15:09:59
Source MAC     : 7c:a9:6b:b1:97:b0
Destination MAC : bc:f1:71:8e:a2:e7
Source IP      : 204.79.197.254
Destination IP  : 192.168.1.34
Protocol       : TCP
Source Port    : 443
Destination Port : 51550
Service        : Unknown
TCP Flags      : RA
Packet Length  : 54 bytes
```

```
Payload:
None
```

=====
Packet #20
=====

Timestamp : 20-07-2026 15:17:53
Source MAC : bc:f1:71:8e:a2:e7
Destination MAC : 7c:a9:6b:b1:97:b0
Source IP : 192.168.1.34
Destination IP : 104.18.20.246
Protocol : TCP
Source Port : 59102
Destination Port : 443
Service : HTTPS
TCP Flags : A
Packet Length : 54 bytes

Payload:
None

Capture Completed Successfully!
CSV File : captured_packets.csv
PCAP File : captured_packets.pcap
Total Packets Captured : 20

Test Cases & Observations

Test	Expected	Result
Capture 20 packets	20 packets displayed	Pass
Display IPs	Source/Destination IP	Pass
Detect TCP	Protocol shown	Pass
CSV Export	CSV created	Pass
PCAP Export	PCAP created	Pass

Future Scope

- GUI using Tkinter/PyQt
- Protocol filtering
- Real-time graphs
- Threat detection
- Wireshark integration

Conclusion

The project successfully meets the CodeAlpha Task 1 requirements and provides practical experience in packet capture and protocol analysis.

References

- Scapy Documentation
- Python Documentation
- CodeAlpha Internship Task Sheet